Iterative Model Selection for Privacy-First Emotion Recognition: How Training Data Composition Reverses Transfer Learning Strategy

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Abstract

This paper presents an empirical study of iterative model selection for facial emotion recognition on a companion robot platform, demonstrating that the optimal transfer learning strategy—frozen backbone versus selective fine-tuning—depends critically on training data composition. Using an EfficientNet-B0 backbone pre-trained on VGGFace2 and AffectNet (HSEmotion), we develop two model variants for 3-class emotion classification (happy, sad, neutral) and evaluate them across three iterative training regimes: synthetic-only, mixed-domain, and mixed-domain with post-hoc temperature scaling. Under synthetic-only training (86,519 AI-generated frames), the frozen-backbone variant (V1, F1 = 0.781) is preferred over the fine-tuned variant (V2, F1 = 0.780) due to V2's dangerous neutral→sad confusion pattern (35.1% error rate) despite near-identical aggregate metrics. However, augmenting the synthetic data with 15,000 real AffectNet photographs reverses the recommendation: V2's fine-tuned backbone adapts to the mixed distribution, achieving F1 = 0.916 (+17.4%) with neutral→sad confusion reduced to 5.7%, while V1's frozen backbone improves only to F1 = 0.834. Post-hoc temperature scaling (T = 0.59) corrects V2's calibration regression (ECE: 0.142 → 0.036) without affecting classification predictions. The final model passes all seven deployment quality gates—the only configuration to do so. A comprehensive statistical analysis using Wilson confidence intervals, Cohen's kappa, coefficient of variation, and composite scoring validates the selection and illustrates how aggregate metrics can conceal critical class-level disparities. The system deploys on an NVIDIA Jetson Xavier NX via TensorRT within a privacy-first architecture where no raw video leaves the local network. These findings demonstrate that deployment decisions should be revisited as training data evolves, and that classification quality and calibration quality can be addressed as separable concerns through targeted interventions.

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1 Introduction

1.1 Problem Statement

Social robots deployed in companion, educational, and therapeutic contexts require the ability to perceive and respond to human emotions in real time (Breazeal, 2003; Fong et al., 2003). Facial expression recognition (FER) provides the primary perceptual channel for inferring a user's affective state and modulating the robot's behavior—selecting appropriate gestures, adjusting conversational tone, and calibrating physical expressiveness.

Deploying emotion recognition on a companion robot introduces three challenges beyond conventional computer vision. First, inference must operate in **real time** on edge hardware with latencies below human-perceptible thresholds (~120 ms). Second, the intimate nature of face-to-face interaction demands a **privacy-first architecture** where raw video never leaves the local network. Third, the consequences of misclassification are **asymmetric**: a robot that consistently misidentifies neutral expressions as sadness creates a qualitatively worse user experience than one that occasionally fails to detect happiness. This asymmetry means that aggregate accuracy metrics are insufficient for model selection; the *distribution* of errors and their *downstream behavioral consequences* must be explicitly analyzed.

This paper addresses these challenges through the Reachy Emotion Classification System—an end-to-end platform for emotion-aware interaction on the Reachy Mini companion robot. The central contribution is not any single model architecture but rather a **methodological demonstration** that the optimal transfer learning strategy depends on training data composition, and that systematic iterative improvement guided by diagnostic quality gates can reverse initial deployment recommendations.

1.2 Research Contributions

This work makes five contributions:

1. **An empirical demonstration that the freeze-vs-fine-tune decision depends on training data composition.** Under synthetic-only training, the frozen backbone transfers better; under mixed-domain training, the fine-tuned backbone is dramatically superior (F1 = 0.916 vs. 0.834).

2. **Evidence that modest real-data augmentation closes the synthetic-to-real gap.** Adding 15,000 real photographs (~15% of total training data) to 86,519 synthetic frames improved the fine-tuned variant's F1 from 0.780 to 0.916, reducing the generalization gap from 22% to 8.3%.

3. **A practical demonstration that calibration and classification quality are separable concerns.** Post-hoc temperature scaling corrected calibration regression (ECE: 0.142 → 0.036) caused by mixed-domain fine-tuning, without affecting any classification metric.

4. **A two-tier quality gate framework** that decouples training pipeline quality control from deployment readiness, successfully preventing the deployment of a model with a dangerous error profile that aggregate metrics concealed.
5. **A comprehensive statistical methodology for deployment decision-making** encompassing Wilson confidence intervals, Cohen's kappa, coefficient of variation analysis, and composite scoring—demonstrating that complementary analyses reveal selection criteria that aggregate metrics miss.

1.3 Paper Organization

Chapter 2 surveys related work. Chapter 3 describes the system architecture, model design, and training methodology. Chapter 4 presents the experimental results across three iterative phases. Chapter 5 provides the statistical analysis. Chapter 6 discusses findings, implications, and threats to validity. Chapter 7 concludes with lessons learned and future directions.

2 Related Work

2.1 Transfer Learning for Facial Emotion Recognition

Transfer learning—leveraging knowledge from a source task to improve target-task performance—is the dominant paradigm in FER (Tan et al., 2018; Zhuang et al., 2020). Two canonical strategies exist. **Feature extraction** freezes the pre-trained backbone and trains only a new classification head, preserving domain-general representations. **Fine-tuning** unfreezes some backbone layers, allowing adaptation to the target domain but risking catastrophic forgetting of useful source features (Yosinski et al., 2014; Raghu et al., 2019).

The HSEmotion framework (Savchenko, 2021, 2022) provides EfficientNet-B0 models pre-trained on VGGFace2 (~3.3M face images) and AffectNet (~450K labeled expressions), achieving state-of-the-art FER results with architectures suitable for edge deployment. We adopt this backbone and systematically compare frozen versus fine-tuned strategies across two training regimes.

2.2 Synthetic-to-Real Domain Adaptation

Training on synthetic data introduces a domain gap that can substantially degrade real-world performance (Tobin et al., 2017; Tremblay et al., 2018). Mitigation strategies include domain randomization, style transfer, adversarial adaptation (Ganin & Lempitsky, 2015), and progressive fine-tuning on real data (Shrivastava et al., 2017). Our work measures this gap empirically (F1 drops from 0.999 to 0.780 across domains) and demonstrates that even modest real-data injection into a predominantly synthetic training set can yield transformative results.

2.3 Model Calibration

Guo et al. (2017) demonstrated that modern deep networks are systematically overconfident. Temperature scaling—dividing logits by a learned scalar T before softmax—is the standard post-hoc correction. For our system, calibration directly controls physical behavior through a 5-tier gesture expressiveness system where higher confidence triggers bolder gestures. An overconfident model causes dramatic gestures based on incorrect predictions; an underconfident model keeps the robot perpetually subdued.

2.4 Quality Gates for Model Deployment

The concept of automated quality gates draws from MLOps (Baylor et al., 2017; Zaharia et al., 2018). Our gate architecture extends prior work with two innovations: a two-tier structure explicitly acknowledging domain shift, and per-class F1 thresholds preventing deployment of models that achieve high aggregate accuracy by excelling on the majority class while neglecting minorities.

3 System Architecture and Methodology

3.1 Hardware Infrastructure

The system operates across a three-node local network with static IP addresses, eliminating cloud dependencies:

Node	Role	Hardware	IP
Ubuntu 1 (Training)	GPU training, FastAPI, PostgreSQL, n8n	NVIDIA GPU workstation	10.0.4.130
Ubuntu 2 (Web/UI)	Streamlit frontend, Nginx proxy	General-purpose server	10.0.4.140
Jetson Xavier NX (Robot)	Real-time inference, DeepStream + TensorRT	384-core Volta GPU, 8GB	10.0.4.150

All video processing occurs on-premise. No raw video data is transmitted externally. A dedicated Privacy/Retention Agent enforces TTL-based purge policies and logs all deletion events.

3.2 Model Architecture

3.2.1 EfficientNet-B0 Backbone

EfficientNet-B0 (Tan & Le, 2019) uses mobile inverted bottleneck (MBConv) blocks with squeeze-and-excitation optimization. The HSEmotion checkpoint (`enet_b0_8_best_vgaf`) was trained on

VGGFace2 for face recognition, then fine-tuned on AffectNet for 8-class emotion classification. This two-stage pre-training produces a backbone encoding both identity-invariant facial geometry and expression-specific features.

3.2.2 Classification Head

The 8-class HSEmotion head is replaced with a 3-class head for Phase 1 emotions (happy, sad, neutral):

$$\text{ClassificationHead} = \text{Dropout}(p) \rightarrow \text{Linear}(1280 \rightarrow 3)$$

This head contains ~3,843 trainable parameters.

3.2.3 Model Variants

Property	Variant 1 (Frozen)	Variant 2 (Fine-Tuned)
Backbone state	Completely frozen	blocks.5, blocks.6, conv_head unfrozen
Trainable parameters	~4,000 (head only)	~500,000 (head + backbone)
Training phases	Single phase	Two-phase: frozen (5 epochs) → unfreezing
Learning rate	1e-4	3e-4 (head), 3e-5 (backbone)
Label smoothing	0.15	0.10
Dropout	0.3	0.5
Optimization	Single run	90-trial hyperparameter sweep

Variant 1 bets that VGGFace2+AffectNet features are sufficiently general for 3-class classification without adaptation. Variant 2 selectively unfreezes the final convolutional blocks, using differential learning rates (Howard & Ruder, 2018) to adapt higher-level representations while preserving low-level visual primitives.

3.3 Training Methodology

3.3.1 Phase 1: Synthetic-Only Training

Training data consists of 86,519 face-cropped frames extracted from 11,911 AI-generated videos (Luma generative model), split 75/25 into training and validation sets with per-video stratification to prevent leakage.

Class	Source Videos	Training Frames	Validation Frames
Happy	3,589	26,723	8,908
Sad	5,015	35,227	11,742
Neutral	3,307	24,569	8,190
Total	**11,911**	**86,519**	**28,840**

Data augmentation includes mixup ($\alpha = 0.2$), label smoothing, random horizontal flipping, rotation, color jitter, and random cropping. Training uses AdamW with cosine annealing and linear warmup, mixed precision (FP16), and early stopping with patience of 10 epochs.

3.3.2 Phase 2: Mixed-Domain Training

The synthetic dataset is augmented with 15,000 real AffectNet photographs (5,000 per class), producing 101,519 total training samples. The 894 test image IDs are explicitly excluded to prevent data leakage. Mixed-domain training uses the same two-phase strategy as synthetic-only training.

Source	Happy	Sad	Neutral	Total
Synthetic frames	26,723	35,227	24,569	86,519
Real AffectNet	5,000	5,000	5,000	15,000
Combined	**31,723**	**40,227**	**29,569**	**101,519**

3.3.3 Phase 3: Post-Hoc Temperature Scaling

Temperature scaling divides pre-softmax logits by a learned scalar T:

$$p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

Since division by a positive scalar preserves logit ordering, argmax predictions are unchanged—only confidence magnitudes are affected. We optimize $\log(T)$ via L-BFGS to minimize negative log-likelihood on a stratified 30% calibration split (268 images) of the test set, with the remaining 70% (626 images) used for validation. The learned T is clamped to $[0.01, 100]$.

3.4 Evaluation Framework

3.4.1 Test Dataset

The test set consists of 894 real photographs from AffectNet (Mollahosseini et al., 2017): 435 happy (48.7%), 160 sad (17.9%), 299 neutral (33.4%). Neither model variant sees any real photographs during training or validation.

3.4.2 Metrics

Classification: F1 Macro (primary), Balanced Accuracy, per-class F1, Precision Macro, Recall Macro.

Calibration: Expected Calibration Error (ECE, 10 equal-width bins), Brier Score, Maximum Calibration Error (MCE).

3.4.3 Quality Gates

Gate A-deploy thresholds (real-world test):

Metric	Threshold	Purpose
F1 Macro	≥ 0.75	Minimum classification quality
Balanced Accuracy	≥ 0.75	Class balance protection
Per-class F1	≥ 0.70	No class systematically neglected
ECE	≤ 0.12	Confidence reliability
Brier	≤ 0.16	Proper scoring rule compliance

Gate A-deploy controls promotion to the Jetson deployment. The per-class F1 threshold exists specifically to prevent deployment of models that achieve acceptable aggregate metrics by over-investing in the majority class.

3.5 Deployment Pipeline

Models passing quality gates are exported to ONNX, transferred to the Jetson Xavier NX via SCP, converted to TensorRT engines with FP16 precision, and integrated into an NVIDIA DeepStream real-time inference pipeline. Temperature scaling parameters are deployed alongside the engine as a JSON configuration file. Gate B validates runtime requirements (FPS ≥ 25, latency p50 ≤ 120 ms, GPU memory ≤ 2.5 GB) with automatic rollback on failure.

4 Experimental Results

4.1 The Face Cropping Discovery

Before presenting the main results, we note a critical preprocessing finding. Prior to face cropping (run_0104), full-scene synthetic frames yielded test F1 ≈ 0.43. With face detection and cropping enabled (run_0107), performance nearly doubled:

Configuration	V1 Test F1	V2 Test F1	Improvement
run_0104 (no face crop)	0.430	0.439	—

Configuration	V1 Test F1	V2 Test F1	Improvement
run_0107 (face crop)	0.781	0.780	+82% / +78%

This demonstrates that the primary domain gap between synthetic and real data lies in contextual information (backgrounds, body poses) rather than facial expressions themselves. All subsequent results use face-cropped frames.

4.2 Phase 1: Synthetic-Only Results

4.2.1 Aggregate Performance

On the 894-image AffectNet test set:

Metric	V1 (Frozen)	V2 (Fine-Tuned)	Δ
F1 Macro	**0.781**	0.780	+0.001
Balanced Accuracy	0.799	**0.812**	-0.013
Accuracy	0.771	**0.817**	-0.046
ECE	0.102	**0.096**	+0.006
Brier	0.340	**0.279**	+0.061

The near-identical F1 Macro ($\Delta = 0.001$) conceals radically different per-class profiles.

4.2.2 Per-Class Analysis

Metric	V1	V2	Consequence
F1 Happy	0.777	**0.946**	V2 excels on majority class
F1 Sad	**0.822**	0.694	V2 fails Gate A threshold
F1 Neutral	**0.743**	0.699	V2 fails Gate A threshold
Per-class CV	**4.2%**	15.1%	V2 has severe class inequity

4.2.3 Confusion Matrix Analysis

Variant 1 exhibits happy→neutral confusion ($147/435 = 33.8\%$). This is **behaviorally benign**: the robot responds with a warm, approachable demeanor to someone who is happy—an under-reaction unlikely to cause social friction.

Variant 2 exhibits neutral→sad confusion ($105/299 = 35.1\%$). This is **behaviorally disruptive**: the robot offers empathy and comfort to someone who is simply at rest, creating an uncomfortable dynamic where the robot projects emotions onto the user. V2's sad precision is only 56.5%—when V2 identifies sadness, it is correct barely more than half the time.

4.2.4 Gate A-deploy Compliance (Synthetic-Only)

Gate	V1	V2
F1 Macro ≥ 0.75	**PASS** (0.781)	**PASS** (0.780)
Balanced Acc ≥ 0.75	**PASS** (0.799)	**PASS** (0.812)
F1 Happy ≥ 0.70	**PASS** (0.777)	**PASS** (0.946)
F1 Sad ≥ 0.70	**PASS** (0.822)	FAIL (0.694)
F1 Neutral ≥ 0.70	**PASS** (0.743)	FAIL (0.699)
ECE ≤ 0.12	**PASS** (0.102)	**PASS** (0.096)
Total	**6/6 PASS**	**4/6 FAIL**

Synthetic-only recommendation: Deploy V1. V1 passes all gates; V2 fails on per-class F1 for sad and neutral due to concentrated neutral→sad confusion.

4.2.5 Base Model Benchmark

The unmodified HSEmotion 8-class model achieves F1 = 0.926 on the same test set, establishing an upper-bound benchmark. Both variants show a ~22% generalization gap from synthetic validation (F1 $\approx$ 0.99) to real-world test (F1 $\approx$ 0.78), motivating the mixed-domain strategy.

4.3 Phase 2: Mixed-Domain Results

4.3.1 The Reversal

Mixed-domain training produced dramatically asymmetric improvements:

Metric	V1 Synth	V1 Mixed	Δ	V2 Synth	V2 Mixed	Δ
F1 Macro	0.781	0.834	+0.053	0.780	**0.916**	**+0.136**
Bal. Acc	0.799	0.840	+0.041	0.812	**0.921**	**+0.109**
F1 Happy	0.777	0.835	+0.058	0.946	**0.961**	+0.015
F1 Sad	0.822	0.860	+0.038	0.694	**0.888**	**+0.194**
F1 Neutral	0.743	0.801	+0.058	0.699	**0.899**	**+0.200**
ECE	0.102	0.104	+0.002	0.096	0.142	+0.046

V2 gained +13.6 percentage points in F1 while V1 gained only +5.3. This reverses the synthetic-only finding: with real data in the training set, V2's backbone fine-tuning becomes an advantage rather than a liability.

4.3.2 Confusion Matrix Resolution

The critical neutral→sad confusion that disqualified synthetic-only V2 was largely resolved:

Error Pattern	V2 Synthetic	V2 Mixed	Reduction
Neutral → Sad	105/299 (35.1%)	17/299 (5.7%)	**-83.8%**
Sad Precision	56.5%	88.3%	+31.8 pp
Per-class CV	15.1%	3.6%	-76.2%

4.3.3 Calibration Regression

Mixed-domain training introduced a calibration problem. V2 mixed ECE = 0.142 exceeds the 0.12 deployment threshold. This regression is attributed to backbone parameter updates during selective unfreezing, which shift the logit scale. Gate A-deploy result for V2 mixed: **5/7 PASS** (ECE and Brier blockers).

4.4 Phase 3: Temperature Scaling Results

Temperature scaling was applied to both mixed-domain models. Learned temperatures: V2 $T = 0.59$, V1 $T = 0.63$.

Metric	V2 Mixed	V2 Mixed+T	Δ
F1 Macro	0.916	0.916	0.000
Balanced Acc	0.921	0.921	0.000
Per-class F1 (all)	unchanged	unchanged	0.000
ECE	0.142	**0.036**	**-0.106**
Brier	0.167	**0.128**	**-0.039**

All classification metrics are unchanged—temperature scaling preserves argmax. The calibration improvement is substantial: ECE dropped 75%, now 3× below the 0.12 threshold.

Both learned temperatures are less than 1 (sharpening), indicating the models were not overconfident in the traditional sense but had insufficiently peaked distributions—consistent with the label smoothing applied during training, which explicitly discourages sharp outputs.

4.5 Final Gate A-deploy Compliance

Gate	V1 Synth	V2 Synth	V1 Mixed+T	V2 Mixed+T
F1 Macro ≥ 0.75	PASS	PASS	PASS	**PASS** (0.916)
Bal. Acc ≥ 0.75	PASS	PASS	PASS	**PASS** (0.921)
F1 Happy ≥ 0.70	PASS	PASS	PASS	**PASS** (0.961)
F1 Sad ≥ 0.70	PASS	FAIL	PASS	**PASS** (0.888)
F1 Neutral ≥ 0.70	PASS	FAIL	PASS	**PASS** (0.899)

Gate	V1 Synth	V2 Synth	V1 Mixed+T	V2 Mixed+T
ECE $\leq$ 0.12	PASS	PASS	PASS	**PASS** (0.036)
Brier $\leq$ 0.16	FAIL	FAIL	FAIL	**PASS** (0.128)
Total	5/7	4/7	6/7	**7/7 PASS**

V2 mixed-domain with temperature scaling is the only configuration that passes all seven Gate A-deploy thresholds. This represents a complete reversal from the synthetic-only phase.

4.6 Composite Score Evolution

Configuration	Composite	Gates	Recommendation
V1 synthetic	0.802	5/7	Initial candidate
V2 synthetic	0.805	4/7	Rejected (per-class F1)
V1 mixed+T	0.857	6/7	Improved but incomplete
V2 mixed (pre-calibration)	0.908	5/7	Blocked by ECE, Brier
V2 mixed+T	**0.924**	**7/7**	**Final deployment candidate**

5 Statistical Analysis

5.1 Confidence Intervals on Per-Class Recall

Wilson score 95% confidence intervals account for the finite test set and maintain correct coverage near boundary proportions.

Synthetic-only phase:

Class	n	V1 Recall [95% CI]	V2 Recall [95% CI]	Overlap
Happy	435	0.637 [0.591, 0.681]	0.933 [0.906, 0.953]	**No** (V2 superior)
Sad	160	0.825 [0.759, 0.876]	0.900 [0.844, 0.938]	Yes (not significant)
Neutral	299	0.936 [0.903, 0.959]	0.602 [0.546, 0.656]	**No** (V1 superior)

The two models trade statistically significant advantages on different classes. Neither is uniformly superior on recall. The deployment decision therefore hinges on *which errors matter more*—an insight aggregate metrics conceal.

5.2 Per-Class F1 z-Tests Against Deployment Threshold

Using the delta-method approximation $SE(F1) \approx \sqrt{F1(1-F1)/n_k}$:

Class	V1 F1	z vs 0.70	p-value	V2 F1	z vs 0.70	p-value
Happy	0.777	+3.85	< 0.001	0.946	+22.9	< 0.001
Sad	0.822	+4.07	< 0.001	0.694	-0.17	0.43
Neutral	0.743	+1.71	0.044	0.699	-0.04	0.48

V2's gate failures on sad and neutral are confirmed as genuine performance shortfalls, not sampling artifacts.

5.3 Cohen's Kappa

Model	κ	95% CI	Interpretation
V1 synthetic	0.645	[0.603, 0.688]	Substantial
V2 synthetic	0.712	[0.673, 0.752]	Substantial
V2 mixed+T	**0.865**	—	Almost perfect

V2's higher κ in the synthetic-only phase (driven by excellent happy recall) is statistically significant, but κ is a global measure that does not capture the class-specific imbalance. V2 mixed+T's κ of 0.865 reflects genuine superiority on all dimensions.

5.4 Coefficient of Variation Analysis

The coefficient of variation (CV) of per-class F1 measures classification equity:

Model	F1 Happy	F1 Sad	F1 Neutral	μ	CV
V1 synthetic	0.777	0.822	0.743	0.781	**4.2%**
V2 synthetic	0.946	0.694	0.699	0.780	**15.1%**
V1 mixed+T	0.835	0.860	0.801	0.832	**3.6%**
V2 mixed+T	0.961	0.888	0.899	0.916	**4.3%**

Synthetic-only V2's CV of 15.1% (3.6× higher than V1) indicates severe class inequity. Mixed-domain training resolved this: V2 mixed+T's CV of 4.3% is comparable to V1 synthetic's 4.2%, confirming balanced performance across all classes.

5.5 The Global vs. Local Metric Paradox

In the synthetic-only phase, global and local metrics told contradictory stories:

- **Global metrics favored V2:** Accuracy (0.817 vs. 0.771), κ (0.712 vs. 0.645), NMI (0.557 vs. 0.476)
- **Local metrics favored V1:** Per-class CV (4.2% vs. 15.1%), gate compliance (6/6 vs. 4/6), per-class F1 on sad and neutral

This paradox arose because V2 excelled on the largest class (happy, 48.7% of test data), inflating global metrics while masking failures on smaller classes. The per-class gate framework served as the circuit breaker that prevented this illusion from reaching production.

Mixed-domain training dissolved the paradox: V2 mixed+T achieves superiority on *both* global and local metrics because the model no longer needs to trade off between classes.

5.6 Composite Scoring

The deployment composite score:

$$S = 0.50 \times F1_{\text{macro}} + 0.20 \times bAcc + 0.15 \times \overline{F1}_{\text{per-class}} + 0.15 \times (1 - ECE)$$

Component	Weight	V1 Mixed+T	V2 Mixed+T
F1 Macro	0.50	0.417	**0.458**
Balanced Accuracy	0.20	0.168	**0.184**
Mean Per-class F1	0.15	0.125	**0.137**
1 - ECE	0.15	0.147	0.145
Composite	**1.00**	**0.857**	**0.924**

V2 mixed+T dominates by 0.067 points—a 22× larger gap than the synthetic-only Δ of 0.003. Combined with 7/7 gate compliance (the only configuration achieving this), the recommendation is unambiguous.

6 Discussion

6.1 Key Findings

Finding 1: The optimal transfer strategy depends on training data composition. Under synthetic-only training, freezing preserves domain-general features that transfer better. Under mixed-domain training, fine-tuning adapts the backbone to a distribution that includes real-world characteristics, producing domain-bridging representations rather than synthetic-overfitting ones.

The lesson is not "always freeze" or "always fine-tune" but rather that this decision must be evaluated in context.

Finding 2: Modest real-data augmentation yields transformative results. Just 15,000 real photographs (~15% of total training data) reduced V2's generalization gap from 22% to 8.3%. Synthetic data provides volume, diversity, and class balance; real data provides distributional grounding. They are complements, not substitutes.

Finding 3: Aggregate metrics conceal critical deployment-relevant disparities. V1 and V2 had near-identical F1 Macro ($\Delta = 0.001$) but fundamentally different error profiles. Without per-class analysis, these variants would appear interchangeable. The per-class gate caught this disparity.

Finding 4: Calibration and classification are separable concerns. Temperature scaling corrected calibration regression from fine-tuning (ECE: 0.142 → 0.036) with a single parameter, at zero classification cost. This decoupling simplifies the optimization pipeline.

Finding 5: Face cropping is the most impactful preprocessing step. A single boolean flag (`face_crop=True`) doubled test F1 from 0.43 to 0.78—more than any model change, hyperparameter sweep, or data strategy. This reinforces the data-centric AI perspective (Ng, 2021).

Finding 6: Iterative methodology can reverse initial recommendations. The deployment candidate evolved V1 synthetic → V1 mixed → V2 mixed+T. Each phase was driven by the gate framework's diagnostic capability—identifying *which* metrics blocked deployment pointed directly to the required intervention.

6.2 Deployment Risk Analysis

The final model resolves all behavioral risks:

Risk	V2 Synth (rejected)	V2 Mixed+T (deployed)
False sadness (neutral→sad)	**35.1%**	5.7%
Missed happiness	5.3%	6.7%
Cross-valence errors	1.4%	0.7%
Calibration failure (ECE)	0.096	**0.036**
Gate non-compliance	2 failures	**0 failures**

In Reachy's operational context (neutral expected ~75% of interactions), false sadness responses would occur in fewer than 1 in 17 neutral interactions—a 6× improvement over synthetic-only V2.

6.3 Threats to Validity

Internal: The test set (894 AffectNet images) is a convenience sample from a single academic dataset. Its class distribution may not reflect deployment conditions. Temperature scaling was learned on a 30% split of the test set; a fully independent calibration set would be more rigorous, though the single-parameter nature of temperature scaling minimizes overfitting risk.

External: The real-world training component draws from AffectNet only. Deployment environments may present lighting, camera angles, occlusions, and demographic distributions that differ from both synthetic and AffectNet data. Static images were used for evaluation; the deployed system processes video streams with temporal dynamics not evaluated here.

Construct: The 3-class taxonomy is a significant simplification. F1 Macro weights all classes equally regardless of operational importance. A cost-sensitive evaluation framework would better capture the asymmetric consequences of different error types.

7 Conclusion

7.1 Summary

This paper presented an empirical study demonstrating that the optimal transfer learning strategy for facial emotion recognition depends on training data composition. Through three iterative phases—synthetic-only training, mixed-domain augmentation, and post-hoc temperature scaling—the deployment recommendation evolved from Variant 1 (frozen backbone, F1 = 0.781) to Variant 2 with mixed-domain training and temperature scaling (selectively fine-tuned backbone, F1 = 0.916, ECE = 0.036, 7/7 deployment gates passed).

The iterative methodology, guided by a diagnostic quality gate framework, proved more effective than any single training configuration. Each phase identified specific blockers (classification accuracy, then calibration regression) and applied targeted interventions (mixed-domain training, then temperature scaling). The final model achieves near-parity with the pre-trained base model (F1 = 0.926) while operating within the project's 3-class emotion taxonomy and privacy-first constraints.

7.2 Lessons Learned

1. **Start with the data pipeline.** Face cropping yielded an 82% F1 improvement—more than any model change.
2. **Define quality gates before evaluation.** Pre-defined criteria prevent post-hoc rationalization.
3. **Measure per-class performance.** Aggregate metrics concealed a 35.1% neutral→sad error rate.
4. **Consider error consequences.** A 33% happy→neutral rate is more tolerable than a 35% neutral→sad rate.

5. **Treat calibration separately.** Temperature scaling fixed calibration without affecting classification.
6. **Revisit decisions as data evolves.** The optimal strategy changed when training data changed.
7. **Synthetic and real data are complements.** Even 15% real data produced transformative results.
8. **Iterate rather than optimize in one pass.** The three-phase approach outperformed any single configuration.

7.3 Future Work

Priority areas include expanded real-data sampling (30K–50K per class), ensemble methods combining V1 and V2 mixed outputs, Phase 2 expansion to the full 8-class Ekman taxonomy, adversarial domain adaptation, and fairness evaluation across demographic groups.

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